

WATERWALKER · CALCULATION NOTE

WaterWalker sizing calculations

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-02	Amish Chadha	Constructable design (WWK-DDR-003); mass, frame load path, rear head tube brackets, cradle bearers and lock pin checked; budget treated as a value-engineering target; all results recalculated
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); 1.2 mm main-tube wall, thinner cradle and parking lock pin applied; all results recalculated
0.1	2026-09-25	Amish Chadha	First TRL 3 sizing note (geometry, mass, push force on firm ground, slopes and sand, assist, brakes, frame, axles, turning, cost, requirement table)

WaterWalker sizing calculations

On paper, the first prototype (no assist, with mounting points) carries 80 L, turns in 3.7 m, lifts containers only 364 mm and has parts estimated at USD 462, USD 12 over the USD 450 value-engineering target. Version 0.3 follows the constructable design of WWK-DDR-003 (open for Amish's review): a cradle carrier on four hangers, two-level rear head tube brackets, headsets and steering locks at all four forks, and the fixings the concept lacked. The carrier now weighs 39.6 kg empty (37.4 kg in v0.2). **Four requirements are not met:** R2 (firm path, 60.8 N against 60 N), R3 (loose sand, 240 to 365 N against 150 N), R4 (10 % climb, 181.5 N against 180 N) and R9 (39.6 kg against 35 kg); R2 and R4 slipped from at risk with the 2.2 kg added for construction. Five are at risk: R1, R6, R7, R10 and R12. The new load path (the rail carried on the hip sleeve, the cradle hung from the cross members) lowers the weld stress range in the rail to 61 MPa, below the 71 MPa reference, and the new rear brackets, cradle bearers and two-tab lock pin all have a factor of 2 or more on yield.

Every number in this note is printed by `docs/04-calcs/sizing.py` (run from the repo root: `python docs/04-calcs/sizing.py`), which also writes `docs/04-calcs/results.csv`. The script reads the geometry from `cad/src/model.py` and the prices from `bom/bom.csv`, so the note, the model, drawing WWK-DWG-001 and the BOM share one source. All values are first-principles estimates; nothing here is measured.

■ 1. Assumptions

Table 1. Inputs. All are assumptions for a paper design.

Input	Value	Basis
Load	Four 20 L jerrycans, 80 kg of water plus 4 x 1.1 kg; or two 20 L clay pots of 8 kg each	WWK-REQ-001 R1
Wheels	26 in, ETRTO 559 rim, 2.1 in (54 mm) tires, radius 333.5 mm; 2.4 in (61 mm) also fits, radius 340.5 mm	Decided, WWK-DDR-001 item 1
Hubs	Front-type 100 mm over-locknut hubs on all four wheels; 70 mm drums on the rear pair	Decided, WWK-DDR-001 item 9 (WWK-DDR-002)
Forks	Four identical rigid 26 in forks, 60 mm offset: the caster trail at the front; the rear head tubes stand 60 mm behind the rear axle	WWK-DDR-003, P3; offset to confirm when bought
Rolling resistance, firm dirt path	Crr 0.02 to 0.05	Typical for wide bicycle tires on packed to rough dirt
Rolling resistance, loose sand	Sinkage 25 to 55 mm, giving Crr 0.197 to 0.300 (section 4)	To be measured in the field
Grade	10 % climb and descent (R4); 20 % parking (R10)	WWK-REQ-001
Steel	Mild steel, yield 235 MPa, 7,850 kg/m ³ , G = 79 GPa	S235 class; ERW tube is often stronger
Dynamic factor	2.5 g on static loads	Ruts, stones and kerbs at walking speed
Target factor on yield	1.5 or more at 2.5 g	Paper design margin
Component masses	Rim 0.55 kg, spokes 0.30, puncture-resistant tire 0.85, thorn-resistant tube 0.45, liner 0.20, drum hub 0.75, plain hub 0.35, rigid fork 1.10, headset 0.12, lock collar and plate 0.06, lock pin 0.03 each	Typical catalogue values for utility parts
Drum brake	70 mm drum, brake factor $C^* = 0.8$, cam ratio 5, lever ratio 4, cable efficiency 0.8, one lever to both drums	Typical cable-operated drums
Assist motor (later)	250 W, 48 V geared hub motor, 40 N·m at low speed; 40 % efficient at 17 to 29 rpm	Assumed; geared hub motors are rated at much higher speeds
SwapCell pack	46.8 V nominal, 452 Wh at minimum cell capacity, 2.85 kg, 20 A continuous, 15 A in legacy mode	SwapCell interface v0.3 and SWC-CAL-001
Household use	100 L per day, 3 km round trip at 4 km/h	WWK-PRB-001

■ 2. Geometry and layout

The layout is set by three limits that squeeze each other: overall width (R6, 900 mm), clear walking width (R7, 600 mm) and 100 mm bicycle hubs held on both sides. With a 770 mm track the carrier is 898 mm wide over the axle nuts and 610 mm clear between the side rails, so both R6 and R7 hold, each by less than 2 %. The overall length is 2,197 mm (2,211 mm with 2.4 in tires).

Two TRL 2 layout faults were found and fixed in the model:

- **Rear axles.** The TRL 2 concept hung each rear wheel on a stub axle outboard of the side rail. A standard M10 bicycle axle cantilevered 40 mm carries about 32.7 N·m at 2.5 g, which is about 472 MPa, beyond what a bicycle axle survives in fatigue. Holding each rear wheel in a fixed rigid bicycle fork with 100 mm dropouts cuts this to about 89 MPa and uses the same fork as the front casters (decided, WWK-DDR-001 item 9, WWK-DDR-002).
- **Caster sweep.** A swivelling front wheel sweeps a circle of 400.5 mm radius about its swivel axis (60 mm trail plus the 2.4 in tire radius). The TRL 2 risers sat inside that circle, so the tire would hit them after only a few degrees of steer. The risers now stand at $x = 1,150$ mm and the wheelbase is 1,530 mm, which leaves 19.5 mm of clearance (decided, WWK-DDR-001 item 10, WWK-DDR-002).

The cradle is 780 x 430 mm outside and, with the 4 mm walls of v0.2, 772 mm long inside. It rests on two bearers 150 mm above the ground, so the ground clearance under the carrier is 125 mm (WWK-DDR-003, P4). Two jerrycans in line (720 mm) leave 48 mm; two 380 mm clay pots leave only 12 mm, so the pot case (R1) is at risk until real pot sizes are known. The heaviest lift is over the side rail at 364 mm (R8).

■ 3. Mass

Table 2. Mass roll-up, first prototype.

Part	Mass (kg)	Basis
Frame, rear brackets and cradle carrier	14.46	9.72 m of tube from <code>frame_members()</code> in the model: 5.24 m of 40 x 30 x 1.2 mm (6.67 kg), 0.93 m of 30 x 30 x 1.5 mm (1.25 kg), 2.88 m of 25 x 25 x 1.5 mm (3.18 kg), 0.67 m of 20 x 20 x 1.5 mm (0.58 kg); head tubes, receiver plate, tabs, pin guides, welds and paint
Rear wheels with drum hubs (2)	6.20	Table 1
Front wheels (2)	5.40	Table 1
Forks (4)	4.40	Table 1
Headsets, lock collars and lock pins (4 sets)	0.84	Table 1 (WWK-DDR-003)
Hip bar, posts, pad and grips	2.20	Estimate
Cradle	3.29	6 mm plywood floor 1.21, 2 mm rubber pad 0.75, 4 mm walls 0.93, straps 0.40
Brake lever, latch and cables	0.45	Estimate
Skirt guards (2)	1.29	3 mm HDPE
Hardware and reflectors	0.60	Estimate
Cradle brackets and bolts; guard clips and bolts	0.33	Estimate (WWK-DDR-003)
Parking lock pin, two tabs and lanyard	0.13	Estimate (WWK-DDR-002, WWK-DDR-003)
Total, empty	39.59 (87.3 lb)	R9 limit 35 kg

The assist kit adds 6.50 kg (motor wheel less the drum hub, disc brake, controller, sensor, class V1 receiver and the 2.85 kg SwapCell pack), for 46.09 kg against the 42 kg limit. Loaded with four jerrycans, the carrier weighs 124.0 kg, or 130.5 kg with the assist kit. The loaded centre of mass is 716 mm ahead of the rear axle and 362 mm above the ground, so the front wheels carry 46.8 % of the load: about 284 N per front wheel and 324 N per rear wheel.

R9 is **not met**, by 4.6 kg. In v0.2 the carrier was 37.35 kg (40.85 kg in v0.1). Making the design constructable (WWK-DDR-003) added 2.24 kg: the cradle carrier and second rear brackets (the 25 x 25 tube rose from 0.89 to 2.88 m), headsets at the rear forks and lock collars at all four, cradle brackets, guard fixings and the second lock pin tab; shorter rails and the removal of the old dropout tabs and the stub on each caster arm saved a little. The two mass options left open in WWK-DDR-002, tires without a puncture belt and no tire liners, would save 1.60 kg (37.99 kg); this is an open decision in WWK-DEC-001.

■ 4. Push force

Push force is $F = m g (\text{Crr} \cos \alpha + \sin \alpha)$ for a grade angle α .

On a firm, level dirt path the loaded carrier needs 24.3 to 60.8 N, which is 24 to 61 W of human power at 1.0 m/s. R2 is **not met**, by 0.8 N at the rough end (59.7 N and at risk in v0.2). Climbing a 10 % grade on firm ground takes 145.2 to 181.5 N; R4 is **not met**, by 1.5 N (178.3 N and at risk in v0.2). Both follow from the mass: with the two open mass options as well the figures would be 60.0 N and 179.2 N, both at risk.

Loose sand. A rigid wheel that sinks z into soil meets it at an entry angle θ , where $\cos \theta = 1 - 2z/D$, and the soil reaction passes through the axle, so the rolling resistance coefficient is $\tan(\theta/2)$. For a 667 mm wheel, a sinkage of 25 to 55 mm gives Crr 0.197 to 0.300, which matches the 0.2 to 0.3 used at TRL 2. The loaded carrier then needs **240 to 365 N**, which is 1.6 to 2.4 times the 150 N target. R3 is **not met**, and the first prototype has no assist.

Table 3. Effect of tire size on the sand push force (loaded, unassisted).

Tire	Crr factor	Sand push force (N)
26 x 2.1 in	1.00	240 to 365
26 x 2.4 in	0.95	228 to 346
26 x 3.0 in	0.86	207 to 314
26 x 4.0 in (fat bike)	0.75	180 to 274
28 x 2.0 in	0.97	233 to 355

The factors scale sinkage with Bekker's pressure-sinkage relation for a soil whose frictional modulus dominates, with a sinkage exponent n of 1.1, typical of dry sand, so sinkage varies as $(W / (b \sqrt{D}))^{2/(2n+1)}$, and Crr as $\sqrt{z/D}$. They are relative only. Even fat-bike tires leave the push force well above 150 N, so the TRL 2 question of whether wider tires could avoid assist is answered: they cannot. Carrying only two jerrycans on a sandy stretch (81.8 kg) still takes 158 to 241 N.

On a 10 % descent the carrier pulls downhill with 61 to 97 N more than rolling resistance absorbs.

■ 5. Assist kit (later prototype)

The assist is not in the first prototype (WWK-DDR-001 item 2); this section checks that the mounting points suit a later kit built to SwapCell interface v0.3.

One 250 W geared hub motor at 40 N·m gives 120 N of thrust at the 333.5 mm wheel radius. On loose sand the user would still push 133 to 264 N, so one motor does not meet R3. Meeting R3 on the loosest sand with the assisted mass takes 234 N of thrust, 78 N·m at the wheels: about 39 N·m from each of two motors, one in each rear wheel. With two motors the user pushes 13 to 144 N. Two motors also cancel the yaw from one-sided drive: one motor makes a 46 N·m yaw moment, which locked casters resist with 30 N of side force (5 % of the front load) but free casters leave to the user as a 72 N difference between the hands.

At walking pace the wheel turns at only 17.2 rpm (0.6 m/s) to 28.6 rpm (1.0 m/s), far below where hub motors are efficient. At an assumed 40 % efficiency, one motor on sand uses 180 W and turns 108 W into heat; two motors at the R3 thrust use 351 W (7.5 A from the 46.8 V pack) and make 210 W of heat, so motor temperature is at risk and needs a thermal check before any assist build. Energy use is about 83 Wh/km with one motor and 162 Wh/km with

two on loose sand, so a SwapCell pack (452 Wh) gives about 5.4 km or 2.8 km of assisted sand. The current is well inside the pack's 15 A legacy-mode limit, so a simple controller without CAN can use the pack through the INTERLOCK coding resistor alone (interface v0.3 item W).

■ 6. Brakes and parking (R10)

Parking the assisted carrier (130.5 kg) on a 20 % grade needs 251 N at the tires, or 41.9 N·m in each rear drum. With the assumed drum and cable ratios that needs about 187 N at the lever when the latch is set (178 N unassisted), which is a hard squeeze for many users. Amish decided on 2026-09-25 to add a positive lock pin and keep the latch for short stops (WWK-DDR-001 item 13, WWK-DDR-002). The 10 mm pin is pushed from the walking space through a tab on the inner fork blade, the left skirt guard, the left rear spokes and a tab on the outer blade, 200 mm above and 60 mm behind the axle, so parking no longer depends on hand force. Pinning one wheel puts the whole parking force through that wheel's spokes and fork; the spokes push sideways on the pin with about the tire force times the wheel radius over the pin radius ($251 \text{ N} \times 333.5 / 200$), about 419 N. Held by the inner tab alone, as in the concept, the pin would bend as a cantilever at 213 MPa, a factor of only 1.1; held by both tabs (WWK-DDR-003) it sees 119 MPa, a factor of 2.0. The spokes still take that load in bending, so the pin fit and spoke damage need a test. Traction is the second limit and the pin does not change it: parked facing downhill, the rear wheels carry only 609 N, so the tires need a friction coefficient of 0.41 (0.35 facing uphill); with only the pinned wheel locked, that wheel alone must supply the force. That holds on firm dirt but not reliably on loose gravel or sand. R10 therefore stays **at risk**.

Holding speed on a 10 % descent takes about 72 N at the lever, and stopping from 1.0 m/s within 1.0 m (0.5 m/s^2) with the assisted mass takes about 124 N. The service brake meets R4's descent clause on paper, but sustained 72 N on long descents is tiring.

■ 7. Frame, axles and hip bar

In the constructable design (WWK-DDR-003) the side rail no longer reaches the rear axle: the rear wheel load comes up the fork into the rear head tube, through its two brackets and down the hip sleeve onto the rail. One side rail is therefore treated as a beam on the hip sleeve (60 mm ahead of the rear axle) and the front tire contact. The cradle hangs from the rear and front lower cross members, which load each rail with 215 N (static) at 332.5 mm and 1,150 mm, plus the rail's share of frame mass. Table 4 gives the peak bending stresses at 2.5 g for the 40 x 30 x 1.2 mm rectangular section ($Z = 1,887 \text{ mm}^3$).

Table 4. Frame stresses at 2.5 g.

Location	Moment, static (N·m)	Moment at 2.5 g (N·m)	Stress (MPa)	Factor on yield
Side rail at the rear cross member	77	192	102	2.30
Side rail at the front riser	87	218	115	2.04
Caster arm root	97	244	129	1.82

The TRL 2 section (30 x 30 x 1.5 mm square) would reach 157 MPa at the caster arm root, a factor of only 1.49, so the main members are 40 x 30 mm (decided, WWK-DDR-001 item 11); the decided 1.2 mm wall (item 12) gives a factor of 1.82, above the 1.5 target. Each caster arm now leaves the upper cross member directly over its riser (WWK-DDR-003, P6), so its root moment goes straight down the riser rather than twisting the cross member. Weld

fatigue is **at risk**: a ± 0.75 g swing about 1 g gives a 61 MPa stress range in the rail at the rear cross member, below the 71 MPa of a typical fillet-welded detail at 2 million cycles (75 MPa at the old hanger position in v0.2). Only a test can confirm the life.

Rear head tube brackets. Each rear head tube sits 60 mm behind the axle, on two 25 x 25 x 1.5 mm brackets 60 mm apart that weld to the hip sleeve. The worst case is parking on 20 % with the lock pin through one wheel: 251 N at that tire, about 195 N·m about the bracket level, or 3,243 N of push and pull in the two brackets. The short stubs then bend at 95 MPa (factor 2.5) and the arms carry 23 MPa in tension or compression. Under the rear wheel load at 2.5 g (809 N, shared by the two brackets) the arms see 41 MPa of bending and 16 MPa of twist, 49 MPa combined (factor 4.8). Below the brackets the hip sleeve carries the parking moment at 126 MPa (factor 1.9, static).

Cradle bearers. Each 25 x 25 x 1.5 mm bearer spans 818 mm between its hangers and carries half the loaded cradle: 1,075 N at 2.5 g, 110 N·m, 105 MPa (factor 2.2).

With one front wheel unloaded in a rut, the frame carries 219 N·m of torsion. If one rail took it all the shear stress would be 82 MPa; shared by both rails the frame twists about 2.06 degrees, so a front wheel stays on the ground over a rut about 28 mm deep before the carrier rests on three wheels.

Each rear wheel sits in a fork with 100 mm dropouts, so its M10 axle carries about 88 MPa at 2.5 g; a cantilevered stub axle, as at TRL 2, would reach 468 MPa. The hip bar uprights (30 x 30 x 1.5 mm sleeves) carry a 400 N push shared between them at 616 mm above the rails: 123 N·m and 80 MPa each, a factor of 2.9.

■ 8. Turning and stability

With the casters trailing tangentially and the 2.4 in tires, a turn about the inner rear wheel sweeps a circle of 3.66 m, and a spin about the rear axle centre (one rear wheel rolling backward) sweeps 3.41 m. R5 is met. The swept front tire clears the risers and rails by 22.5 mm at any steer angle (`cad/src/model.py --check`).

The static side tip angle, loaded, is 47 degrees (half track 385 mm, centre of mass 362 mm). The static tip angles over the front and rear axles are 66 and 63 degrees. These are static angles; a wheel dropping into a rut on a cross-slope still needs a dynamic check in the field.

■ 9. Water delivered and cost

A household using 100 L per day needs 5 head-carried trips or 2 WaterWalker trips (1.25 rounded up). On a 3 km round trip at 4 km/h that is 3.75 h against 1.50 h of walking, if walking speed is similar; sandy routes are slower.

Value-engineering target: USD 450 (`budget_usd` , a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 462.00 for the first prototype without assist and with mounting points (USD 12.00 over the target). Making the design constructable added USD 36: four headset and steering lock sets (BOM line 16), the cradle carrier and second rear brackets, cradle brackets, guard clips and fixings. The assist kit (hub motor, sensor and class V1 receiver) adds USD 260.00; the SwapCell pack is priced once in the SwapCell BOM and excluded here. Cost drivers and savings worth trying are in WWK-DEC-001.

■ 10. Results against every requirement

Status rule: **met** means the value is within 95 % of the limit; **at risk** means within the limit but with less than 5 % margin, or dependent on an unverified assumption; **not met** means past the limit. R11 is reported against the value-engineering target.

Table 5. Requirement results (WWK-REQ-001 v0.5).

ID	Target	Value	Status
R1	80 L in four jerrycans or 40 L in two clay pots; payload 90 kg or less	84.4 kg with jerrycans, 56 kg with pots; spare length 48 mm (jerrycans), 12 mm (pots)	At risk (pot size)
R2	60 N or less, firm level path	24.3 to 60.8 N	Not met (by 0.8 N)
R3	150 N or less, loose sand (with assist if fitted)	240 to 365 N unassisted; 133 to 264 N with one motor; 13 to 144 N with two	Not met (first prototype has no assist)
R4	180 N or less on a 10 % climb; controlled descent	145.2 to 181.5 N climb; 72 N at the lever to hold on the descent	Not met (climb, by 1.5 N)
R5	4.0 m turning circle or less	3.66 m about the inner rear wheel; 3.41 m spin	Met
R6	900 mm wide or less	898 mm over the axle nuts; the lock pin ends inside them	At risk
R7	No step-over; walking width 600 mm or more; hip bar 850 to 1,050 mm; skirt guards	0 mm; 610 mm; 850 to 1,050 mm with 150 mm of post in the sleeve at full height; guards on both rear wheels	At risk (walking width margin 10 mm)
R8	Lift 450 mm or less, from either side	364 mm	Met
R9	Empty 35 kg or less (42 kg with assist)	39.6 kg (46.1 kg)	Not met
R10	Service brake while walking; park rated gross mass on 20 %	Lever on the grip; positive lock pin held at both ends for parking (latch 187 N for short stops); tire friction 0.41 needed	At risk
R11	First prototype (no assist, with mounting points): value-engineering target USD 450	USD 462	Over the target by USD 12
R12	Wear parts are standard 26 in bicycle parts	26 in rims, tires, tubes, spokes, cables, 1 1/8 in headsets; 100 mm drum hubs are less common than rim-brake hubs in rural markets	At risk
R13	Assist-ready to SwapCell interface v0.3	Receiver plate, torque-arm tab on the right rear fork and sensor tab in the model; 7.5 A peak against the 15 A legacy limit; latch class V1 retention needs a test	Not verifiable at TRL 3

Summary: 2 met, 5 at risk, 4 not met (R2, R3, R4, R9), R11 over the value-engineering target, 1 not verifiable at TRL 3. In v0.2: 3 met, 7 at risk, 2 not met (R3, R9).

■ 11. Corrections to earlier figures

The TRL 2 figures in WWK-PRC-001 v0.2 were checked against this script and corrected in v0.3 of the precis: empty mass 31 kg to 40.8 kg (47.3 kg with assist); loaded mass 115 kg to 125.2 kg; firm push force 25 to 55 N to 24.6 to 61.4 N; sand push force 225 to 340 N to 242 to 368 N; 10 % climb 135 to 170 N to 146.7 to 183.4 N; hold-back on the descent 55 to 90 N to 61 to 98 N; parking force 250 N to 253 N; turning circle 3.8 m to 3.66 m; width 860 mm to

898 mm; walking width 660 mm to 610 mm; lift height 350 mm to 364 mm; side tip angle 45 degrees to 47 degrees; parts cost about USD 300 to USD 430; assist energy 70 Wh/km to 83 Wh/km; and the assist pack is 46.8 V (SwapCell 13S), not 36 V.

Version 0.2 (WWK-DDR-002) then changed: empty mass 40.8 kg to 37.4 kg; loaded mass 125.2 kg to 121.8 kg; firm push force 61.4 N to 59.7 N; 10 % climb 183.4 N to 178.3 N; sand 242 to 368 N to 236 to 358 N; caster arm stress 112 MPa to 133 MPa; weld stress range 63 MPa to 75 MPa; parts cost USD 430 to USD 426.

Version 0.3 (WWK-DDR-003, constructable design) changed: empty mass 37.4 kg to 39.6 kg; loaded mass 121.8 kg to 124.0 kg; firm push force 59.7 N to 60.8 N; 10 % climb 178.3 N to 181.5 N; sand 236 to 358 N to 240 to 365 N; the rail is carried on the hip sleeve and the cradle hangs from the cross members, so the caster arm stress is 129 MPa and the weld stress range 61 MPa; parts cost USD 426 to USD 462.

Safety: These are paper calculations. The brake, parking and frame figures depend on assumed drum ratios, tire friction and a 2.5 g dynamic factor, none of which has been measured. Nothing in this note makes the carrier safe to build or load. TRL 4 testing is on hold by Amish's instruction.